

Second Terminal Examination - 2079

Class : XI

Subject: Biology (Code: 2011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

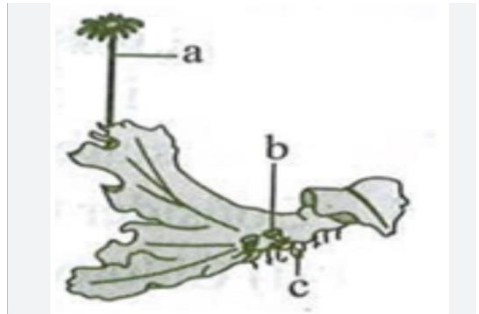
PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet.

[6 × 1 = 6]

1. The food chain which starts from dead & decaying organic matter is called
 - a. Detritus food chain
 - b. Predatory food chain
 - c. Grazing food chain
 - d. Parasitic food chain
2. The functional unit of mitochondria are called
 - a. Cristae
 - b. Oxyosomes
 - c. Matrix
 - d. Inner membrane
3. Scalariform conjugation takes place between
 - a. Two opposite strains of adjacent cells of same filament
 - b. Two opposite strains of three filaments
 - c. Two opposite strains of different filament
 - d. Two same strains of adjacent cells of same filament
4. The labelled part 'a' shown in the diagram below is
 - a. Antheridium
 - b. Archegonium
 - c. Gemma cup
 - d. Archegoniophore



5. The two major aspects of an ecosystem are
 - a. Ecological pyramid & nutrient cycling
 - b. food chain & food Web
 - c. Structure & function
 - d. Trophic level & food chain.
6. The taxonomic group present below the class is

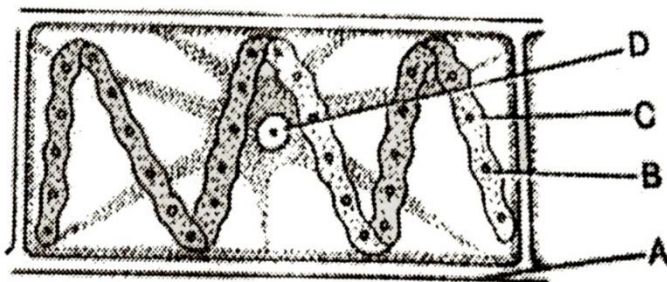
a. Order	b. Division
c. Family	d. Phylum

Group 'B' [BOTANY]

Give short answer to the following questions.

[4 × 4 = 16]

1. A single cell of *Spirogyra* under high power is shown in the diagram below, on the basis of the diagram answer the following questions [1×4 = 4]



- i) What makes the *Spirogyra* filament slippery to touch?
 - ii) Label the parts A & D.
 - iii) How nucleus is suspended in its position?
2. Describe the structure and functions of endoplasmic reticulum. [2+2]
3. Define ecosystem & describe the structural aspects of pond ecosystem [1+3]

OR

- Explain the stratification of pond based on sunlight penetration. [4]
4. Define cell membrane. Write down the functions of cell membrane. [1+3]

Group 'C' [BOTANY]

Give long answer to the following questions.

[2 × 8 = 16]

5. Describe the structural & functional aspects of forest ecosystem in detail. [4+4]
6. Describe the family Brassicaceae in semi-technical terms with floral formula and floral diagram. State the scientific names of any two economic plants of the family. [4+3+1]

OR

Why bryophytes are called amphibians of plant Kingdom? Describe the life cycle of *Marchantia* in detail. [1+7]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet.

[5 × 1 = 5]

1. The study of interaction of antigens and antibodies in the blood is known as:
a. Serology
b. Hematology
c. Angiology
d. Cryobiology
2. In *Paramecium*, osmoregulation occurs through
a. Body surface
b. Cilia
c. Nucleus
d. Contractile vacuole
3. The organs which have same basic structure and origin but differ in function are called as:
a. Homologous organs
b. Analogous organs
c. Vestigial organs
d. None
4. The inheritance of acquired character is given by the concept of:
a. Darwinism
b. Lamarckism
c. Mendelism
d. Neo Lamarckism
5. Phylum Protozoa is classified on base of
a. size
b. shape
c. number of nuclei
d. locomotory organ

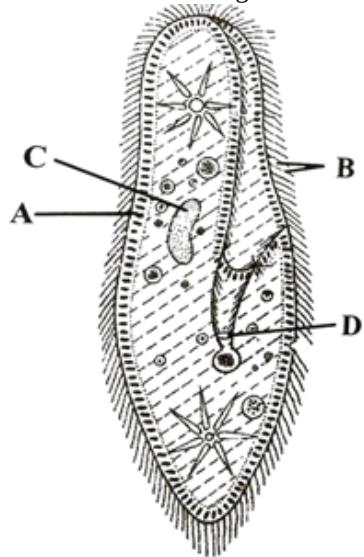
Group 'B' [ZOOLOGY]

Give short answers to the following questions.

[4 × 4 = 16]

1. What are vestigial organs? List 2 vestigial organs present in human body. How vestigial organs support organic evolution? [1+1+2]
2. What causes the emergence of new species? Explain using Modern Synthetic Theory as a foundation. [2+2+4]
3. The following is the diagram of *Paramecium* & answer the following:

- a. Name C & D. [2]
- b. What is the function of A and D. [2]



4. “Ontogeny recapitulates Phylogeny”. Explain this theory. [4]

Group 'C' [ZOOLOGY]

Give long answers to the following questions.

[2 × 8 = 16]

5. What is evolution? Briefly summarize the evolutionary history of modern man starting from Dryopithecus. [1+7]
6. What is conjugation? Describe conjugation in *Paramecium caudatum* with diagrams. Mention any two significances of conjugation. [1+4+2+1]

❧ The End ❧